Chapter 3. Sampling Distributions and Their Approximations

3.1 Population and Sample

Definition 3.1.1 (Def 3.1) — population, sample

- **population**: the totality of objects under study
- **sample**: the part of the population actually observed

Definition 3.1.2 (Def 3.2) — random sample

$X_1, \dots, X_n$ is a random sample of size n from population density f if the joint density factors into the product of f.

$$f_{X_1,\dots,X_n}(x_1, \dots, x_n) = f(x_1)f(x_2)\cdots f(x_n) = \prod_{i=1}^{n} f(x_i)$$

That is, the X_i are iid (independent and identically distributed).

Example 3.1.1 (Ex 3.1) — Bernoulli population

$f(x) = p^x q^{1-x}$, $x = 0,1$, $p + q = 1$, $n = 10$:

$$f_{X_1,\dots,X_{10}}(x_1, \dots, x_{10}) = \prod_{i=1}^{10} p^{x_i} q^{1-x_i} = p^{\Sigma x_i} q^{10-\Sigma x_i} \quad \text{(by independence)}$$

Definition 3.1.3 (Def 3.3) — statistic

A statistic is a function $T = T(X_1, \dots, X_n)$ of the random sample that contains no unknown parameter. Being a function of random variables, T is itself a random variable.

Example 3.1.2 (Ex 3.2) — identifying a statistic

A function that depends on an unknown parameter θ is not a statistic.

- statistic: $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$, $\quad \max\{X_1, \dots, X_n\}$
- not a statistic: $\overline{X}_n - \theta$, $\quad \max\{X_1/\theta, \dots, X_n/\theta\}$

Definition 3.1.4 (Def 3.4) — sample moments

- r-th **sample moment**: $m_r' = \frac{1}{n}\sum_{i=1}^{n} X_i^r$
- r-th **central sample moment**: $m_r = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}_n)^r$

Definition 3.1.5 — sample mean, sample variance

- **sample mean**: $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \ (= m_1')$
- **sample variance**: $S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2$

Unlike m_2, the divisor is $n-1$ rather than n; this makes $E(S_n^2) = \sigma^2$, so S_n^2 is an **unbiased** estimator of σ^2.

Theorem 3.1.1 (Thm 3.1) — moments of $\overline{X}_n$ and S_n^2

For a random sample from a population with mean μ, variance σ^2, and 4th central moment $\mu_4 = E(X_i - \mu)^4$:

1. $E(\overline{X}_n) = \mu$
2. $\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}$
3. $E(S_n^2) = \sigma^2$
4. $\mathrm{Var}(S_n^2) = \frac{1}{n}\left(\mu_4 - \frac{n-3}{n-1}\sigma^4\right), \quad n > 1$

Proof. **(1)**

$$E(\overline{X}_n) = \frac{1}{n}\sum_{i=1}^{n} E(X_i) = \frac{1}{n} n\mu = \mu \quad \text{(by linearity of } E\text{)}$$

(2)

$$\mathrm{Var}(\overline{X}_n) = \frac{1}{n^2}\mathrm{Var}\Big(\sum_{i=1}^{n} X_i\Big) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{\sigma^2}{n} \quad \text{(by independence)}$$

(3) Start from the identity

$$\sum_{i=1}^{n}(X_i - \overline{X}_n)^2 = \sum_{i=1}^{n}(X_i - \mu)^2 - n(\overline{X}_n - \mu)^2$$

Taking expectations on both sides,

$$E[\sum_{i=1}^{n}(X_i-\overline{X}_n)^2]=n\sigma^2-n\cdot\frac{\sigma^2}{n}=(n-1)\sigma^2 \quad (\text{by } E(X_i-\mu)^2=\sigma^2,\ E(\overline{X}_n-\mu)^2 = \mathrm{Var}(\overline{X}_n))$$

$$\therefore\ E(S_n^2)=\frac{1}{n-1}(n-1)\sigma^2=\sigma^2$$

(4) Put $Y_i=X_i-\mu$, so $E(Y_i)=0,\ E(Y_i^2)=\sigma^2,\ E(Y_i^4)=\mu_4$. Since the Y_i are iid, any term in which a distinct index survives with exponent 1 has expectation 0.

$$W:=(n-1)S_n^2=\sum_{i=1}^{n}Y_i^2-n\overline{Y}_n^2 \quad (\text{by the identity in (3)}), \qquad E(W)=(n-1)\sigma^2$$

$$\mathrm{Var}(S_n^2)=\frac{\mathrm{Var}(W)}{(n-1)^2}=\frac{E(W^2)-[E(W)]^2}{(n-1)^2}=\frac{E(W^2)}{(n-1)^2}-\sigma^4$$

$$W^2=(\sum_{i=1}^{n}Y_i^2)^2-2n(\sum_{i=1}^{n}Y_i^2)\overline{Y}_n^2+n^2\overline{Y}_n^4$$

(i) In $E[(\sum_i Y_i^2)^2]=\sum_i\sum_j E(Y_i^2Y_j^2)$, the n terms with $i=j$ give μ_4 and the $n(n-1)$ terms with $i\neq j$ give σ^4 (by independence):

$$E[(\sum_{i=1}^{n}Y_i^2)^2]=n\mu_4+n(n-1)\sigma^4$$

(ii) Substituting $\overline{Y}_n^2=\frac{1}{n^2}\sum_{j,k}Y_jY_k$, only the terms with $j=k$ survive (by independence, $E(Y_j)=0$):

$$E[(\sum_{i=1}^{n}Y_i^2)\overline{Y}_n^2]=\frac{1}{n^2}[n\mu_4+n(n-1)\sigma^4]=\frac{\mu_4+(n-1)\sigma^4}{n}$$

(iii) In $\overline{Y}_n^4=\frac{1}{n^4}\sum_{i,j,k,l}Y_iY_jY_kY_l$, only two cases survive: all four indices equal (n terms, μ_4) and two matched pairs ($3n(n-1)$ terms, σ^4):

$$E(\overline{Y}_n^4)=\frac{1}{n^4}[n\mu_4+3n(n-1)\sigma^4]=\frac{\mu_4+3(n-1)\sigma^4}{n^3}$$

Substituting (i), (ii), (iii),

$$E(W^2) = [n\mu_4 + n(n-1)\sigma^4] - 2n \cdot \frac{\mu_4 + (n-1)\sigma^4}{n} + n^2 \cdot \frac{\mu_4 + 3(n-1)\sigma^4}{n^3}$$

Collecting the coefficients of μ_4 and σ^4,

- coefficient of μ_4: $n - 2 + \frac{1}{n} = \frac{(n-1)^2}{n}$
- coefficient of σ^4: $n(n-1) - 2(n-1) + \frac{3(n-1)}{n} = \frac{(n-1)(n^2-2n+3)}{n}$

$$E(W^2) = \frac{(n-1)^2}{n}\mu_4 + \frac{(n-1)(n^2-2n+3)}{n}\sigma^4$$

Dividing by $(n-1)^2$, subtracting σ^4, and putting the σ^4 term over $n(n-1)$,

$$\frac{n^2-2n+3}{n(n-1)} - 1 = \frac{(n^2-2n+3)-(n^2-n)}{n(n-1)} = -\frac{n-3}{n(n-1)}$$

$$\therefore\ \mathrm{Var}(S_n^2) = \frac{E(W^2)}{(n-1)^2} - \sigma^4 = \frac{\mu_4}{n} - \frac{n-3}{n(n-1)}\sigma^4 = \frac{1}{n}\left(\mu_4 - \frac{n-3}{n-1}\sigma^4\right) \qquad \blacksquare$$

3.2 Distributions Related to the Normal

3.2.1 Chi-squared distribution

Definition 3.2.1 — chi-squared distribution

The gamma distribution with parameters $(k, \theta) = (n/2, 2)$ is called the chi-squared distribution with n degrees of freedom, written $X \sim \chi^2(n)$.

$$f_X(x) = \frac{1}{\Gamma(n/2)}\left(\frac{1}{2}\right)^{n/2} x^{n/2-1} e^{-x/2}, \qquad x > 0 \quad (3.1)$$

Theorem 3.2.1 (Thm 3.2) — mgf, mean, variance of $\chi^2(n)$

1. $M_X(t) = (1-2t)^{-n/2}, \quad t < \frac{1}{2}$
2. $E(X) = n$
3. $\mathrm{Var}(X) = 2n$

Proof. $\chi^2(n) = \text{GAM}(n/2,2)$ and the gamma mgf is $(1-\theta t)^{-k}$, so putting $k = n/2,\ \theta = 2$,

$$M_X(t) = (1-2t)^{-n/2} \quad \text{(by the gamma mgf)}$$

$$E(X) = k\theta = \frac{n}{2}\cdot 2 = n, \qquad \text{Var}(X) = k\theta^2 = \frac{n}{2}\cdot 2^2$$
$$= 2n \quad \text{(by the gamma mean and variance)} \qquad \blacksquare$$

Theorem 3.2.2 (Thm 3.3) — additivity of the chi-squared distribution

For independent $X_i \sim \chi^2(k_i)$, $i = 1, \dots, n$:

$$Y = \sum_{i=1}^{n} X_i \sim \chi^2(\sum_{i=1}^{n} k_i)$$

Proof.

$$M_Y(t) = \prod_{i=1}^{n} (1-2t)^{-k_i/2}$$
$$= (1-2t)^{-\sum_i k_i/2} \quad \text{(by the mgf of an independent sum, Theorem 3.2.1)}$$

This is the mgf of $\chi^2(\sum_i k_i)$, hence the claim (by uniqueness of the mgf). ■

Theorem 3.2.3 (Thm 3.4) — square of a standard normal

$Z \sim N(0,1) \ \Rightarrow\ Y = Z^2 \sim \chi^2(1)$.

Proof.

$$M_{Z^2}(t) = E(e^{tZ^2}) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{tz^2 - z^2/2} dz = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2(1-2t)/2} dz = (1-2t)^{-1/2}, \quad t < \frac{1}{2}$$

(by the Gaussian integral). This is the mgf of $\chi^2(1)$, hence the claim (by uniqueness of the mgf). ■

Theorem 3.2.4 (Thm 3.5) — sum of squared standardized normals

For independent $X_i \sim N(\mu_i, \sigma_i^2)$, $i = 1, \dots, k$:

$$V = \sum_{i=1}^{k} \left(\frac{X_i - \mu_i}{\sigma_i}\right)^2 \sim \chi^2(k)$$

Proof. The standardized variables $(X_i - \mu_i)/\sigma_i \sim N(0,1)$ are independent, each square is $\chi^2(1)$ (by Theorem 3.2.3), and the sum is $\chi^2(k)$ (by Theorem 3.2.2). ■

Corollary 3.2.1 For a random sample from $N(\mu, \sigma^2)$:

$$\sum_{i=1}^{n} \frac{(X_i - \mu)^2}{\sigma^2} \sim \chi^2(n)$$

3.2.2 F distribution

Definition 3.2.2 — F distribution

For independent $U \sim \chi^2(n)$ and $V \sim \chi^2(m)$, the distribution of

$$X = \frac{U/n}{V/m} \quad (3.2)$$

is the F distribution with degrees of freedom (n, m), written $X \sim F(n, m)$, with density

$$f_X(x) = \frac{\Gamma[(n+m)/2]}{\Gamma(n/2)\Gamma(m/2)} \left(\frac{n}{m}\right)^{n/2} \frac{x^{(n-2)/2}}{[1+nx/m]^{(n+m)/2}}, \qquad x > 0 \quad (3.3)$$

Derivation of (3.3). Bivariate change of variables: for a one-to-one map with inverse $u = g(x, y)$, $v = h(x, y)$,

$$f_{X,Y}(x, y) = f_{U,V}(g(x, y), h(x, y))|J|, \qquad J = \frac{\partial(u, v)}{\partial(x, y)}$$

(1) Joint density of (U, V), the product of two densities (3.1) (by independence):

$$f_{U,V}(u, v) = \frac{1}{\Gamma(n/2)\Gamma(m/2)2^{(n+m)/2}} u^{n/2-1} v^{m/2-1} e^{-(u+v)/2}, \qquad u > 0,\ v > 0$$

(2) Attach the auxiliary variable $Y = V$; the inverse map and Jacobian are

$$u = \frac{n}{m}xy, \quad v = y; \qquad J = \left| \begin{matrix} \frac{n}{m}y & \frac{n}{m}x \\ \text{[4pt]}0 & 1 \end{matrix} \right| = \frac{n}{m}y$$

$$f_{X,Y}(x, y) = \frac{(n/m)^{n/2}}{\Gamma(n/2)\Gamma(m/2)2^{(n+m)/2}} x^{n/2-1} y^{(n+m)/2-1} \exp\left[-\frac{y}{2}(1 + \frac{nx}{m})\right]$$

(exponent of y: $(n/2-1)+(m/2-1)+1=(n+m)/2-1$)

(3) Marginalize over y with $a=\frac{1}{2}(1+nx/m)$:

$$\int_0^\infty y^{(n+m)/2-1}\,e^{-ay}dy=\frac{\Gamma[(n+m)/2]}{a^{(n+m)/2}}$$

$$=\Gamma[(n+m)/2]\cdot\frac{2^{(n+m)/2}}{(1+nx/m)^{(n+m)/2}}\quad\text{(by the gamma integral)}$$

The factor $2^{(n+m)/2}$ cancels, giving (3.3). ■

Theorem 3.2.5 (Thm 3.6) — mean and variance of $F(n,m)$

1. $E(X)=\frac{m}{m-2},\quad m>2$
2. $\mathrm{Var}(X)=\frac{2m^2(n+m-2)}{n(m-2)^2(m-4)},\quad m>4$

Proof. $X=\frac{m}{n}\cdot\frac{U}{V}$, where U and $1/V$ are independent and $E(1/V)=1/(m-2)$ for $m>2$, so

$$E(X)=\frac{m}{n}E(U)E(\frac{1}{V})=\frac{m}{n}\cdot n\cdot\frac{1}{m-2}=\frac{m}{m-2}\quad\text{(by the gamma integral)}$$

For the variance, $X^2=\frac{m^2}{n^2}\cdot\frac{U^2}{V^2}$ and

$$E(U^2)=\mathrm{Var}(U)+[E(U)]^2=2n+n^2=n(n+2)\quad\text{(by Theorem 3.2.1)}$$

$$E(\frac{1}{V^2})=\int_0^\infty\frac{1}{v^2}\cdot\frac{v^{m/2-1}e^{-v/2}}{\Gamma(m/2)2^{m/2}}dv=\frac{\Gamma(m/2-2)2^{m/2-2}}{\Gamma(m/2)2^{m/2}}=\frac{1}{(m-2)(m-4)},\quad m>4$$

(by the gamma integral and $\Gamma(m/2)=(m/2-1)(m/2-2)\Gamma(m/2-2)$)

$$E(X^2)=\frac{m^2}{n^2}E(U^2)E(\frac{1}{V^2})=\frac{m^2(n+2)}{n(m-2)(m-4)}\quad\text{(by independence)}$$

$$\mathrm{Var}(X)=E(X^2)-[E(X)]^2=\frac{m^2(n+2)}{n(m-2)(m-4)}-\frac{m^2}{(m-2)^2}=\frac{2m^2(n+m-2)}{n(m-2)^2(m-4)}$$

(numerator: $(n+2)(m-2)-n(m-4)=2(n+m-2)$) ■

Theorem 3.2.6 — reciprocal of an F variable, percentile relation

$X \sim F(n,m) \Rightarrow 1/X \sim F(m,n)$, since $1/X = \frac{V/m}{U/n}$. Hence for upper percentiles

$$\frac{1}{F_\alpha(n,m)} = F_{1-\alpha}(m,n) \quad (3.4)$$

so the lower part of the F table need not be tabulated.

Example 3.2.1 (Ex 3.3) — F percentile

$X \sim F(5,10)$ and $P(X \le a) = 0.01$:

$$a = F_{0.99}(5,10) = \frac{1}{F_{0.01}(10,5)} = \frac{1}{10.1} = 0.099 \quad \text{(by (3.4) and the F table)}$$

3.2.3 t distribution

Definition 3.2.3 — t distribution

For $Z \sim N(0,1)$ and independent $U \sim \chi^2(k)$, the distribution of

$$X = \frac{Z}{\sqrt{U/k}} \quad (3.5)$$

is the t distribution with k degrees of freedom, written $X \sim t(k)$, with density

$$f_X(x) = \frac{\Gamma[(k+1)/2]}{\Gamma(k/2)\sqrt{k\pi}} \cdot \frac{1}{(1+x^2/k)^{(k+1)/2}}, \qquad -\infty < x < \infty \quad (3.6)$$

Derivation of (3.6). The same bivariate change of variables, with auxiliary variable $W = U$.

(1) Joint density (by independence):

$$f_{Z,U}(z,u) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \cdot \frac{1}{\Gamma(k/2)2^{k/2}} u^{k/2-1} e^{-u/2}, \qquad u > 0$$

(2) Inverse map and Jacobian:

$$z = x\sqrt{\frac{w}{k}}, \quad u = w; \qquad J = \begin{vmatrix} \sqrt{\frac{w}{k}} & \frac{x}{2\sqrt{kw}} \\[4pt] 0 & 1 \end{vmatrix} = \sqrt{\frac{w}{k}}$$

$$f_{X,W}(x,w) = \frac{1}{\sqrt{k\pi}\Gamma(k/2)2^{(k+1)/2}} w^{(k+1)/2-1} \exp\left[-\frac{w}{2}(1+\frac{x^2}{k})\right]$$

(exponent of w: $w^{k/2-1} \cdot w^{1/2} = w^{(k+1)/2-1}$; constants: $\sqrt{2\pi} \cdot 2^{k/2} \cdot \sqrt{k} = 2^{(k+1)/2}\sqrt{k\pi}$)

(3) Marginalize over w with $a = \frac{1}{2}(1 + x^2/k)$:

$$\int_0^\infty w^{(k+1)/2-1} e^{-aw} dw = \Gamma[(k+1)/2] \cdot \frac{2^{(k+1)/2}}{(1+x^2/k)^{(k+1)/2}} \quad \text{(by the gamma integral)}$$

The factor $2^{(k+1)/2}$ cancels, giving (3.6). ■

Example 3.2.2 (Ex 3.4) — symmetry of the t distribution

$X \sim t(10)$ with $P(X \leq 1.812) = 0.95$ from the table; the density is symmetric about $x = 0$, so

$$P(X \leq -1.812) = 1 - P(X \leq 1.812) = 0.05$$

Theorem 3.2.7 (Thm 3.7) — mean and variance of $t(n)$

1. $E(X) = 0, \quad n > 1$
2. $\mathrm{Var}(X) = \frac{n}{n-2}, \quad n > 2$

Proof. Z is symmetric about 0 and independent of U, so X is symmetric about 0; for $n > 1$, $E|X| < \infty$ and hence $E(X) = 0$. Then $\mathrm{Var}(X) = E(X^2)$ and

$$X^2 = \frac{Z^2}{U/n} = nZ^2 \cdot \frac{1}{U}, \qquad E(X^2) = nE(Z^2)E(\frac{1}{U}) \quad \text{(by independence)}$$

$$E(Z^2) = \mathrm{Var}(Z) + [E(Z)]^2 = 1$$

$$E(\frac{1}{U}) = \frac{1}{\Gamma(n/2)2^{n/2}} \int_0^\infty u^{(n/2-1)-1} e^{-u/2} du = \frac{\Gamma(n/2-1)2^{n/2-1}}{\Gamma(n/2)2^{n/2}} = \frac{1}{n-2}, \quad n > 2$$

(by the gamma integral $\int_0^\infty u^{a-1} e^{-u/2} du = \Gamma(a)2^a$ and $\Gamma(n/2) = (n/2-1)\Gamma(n/2-1)$)

$$\therefore \ \mathrm{Var}(X) = E(X^2) = n \cdot 1 \cdot \frac{1}{n-2} = \frac{n}{n-2} \qquad ■$$

Corollary 3.2.2 — square of a t variable

$X \sim t(n) \Rightarrow X^2 \sim F(1, n)$.

Proof.

$$X^2 = \frac{Z^2}{U/n} = \frac{Z^2/1}{U/n}$$

with $Z^2 \sim \chi^2(1)$ (by Theorem 3.2.3) independent of $U \sim \chi^2(n)$, which is exactly the definition (3.2) of $F(1, n)$. ■

Notation 3.2.1 — upper percentiles

The point with upper-tail probability α is written as follows.

- z_α: $P(Z > z_\alpha) = \alpha$ for $Z \sim N(0,1)$
- $t_\alpha(n)$: $P(T > t_\alpha(n)) = \alpha$ for $T \sim t(n)$
- $\chi^2_\alpha(n)$: $P(X > \chi^2_\alpha(n)) = \alpha$ for $X \sim \chi^2(n)$
- $F_\alpha(n, m)$: $P(F > F_\alpha(n, m)) = \alpha$ for $F \sim F(n, m)$

$N(0,1)$ and $t(n)$ are symmetric about 0, hence

$$z_{1-\alpha} = -z_\alpha, \qquad t_{1-\alpha}(n) = -t_\alpha(n)$$

3.3 Sampling from a Normal Distribution

Theorem 3.3.1 (Thm 3.8) — sum of independent normals

For independent $X_i \sim N(\mu_i, \sigma_i^2)$:

$$\sum_{i=1}^n X_i \sim N(\sum_{i=1}^n \mu_i\,,\ \sum_{i=1}^n \sigma_i^2)$$

Proof. The normal mgf is $\exp(\mu_i t + \sigma_i^2 t^2/2)$, so

$$\begin{aligned} M(t) &= \prod_{i=1}^n \exp(\mu_i t + \frac{\sigma_i^2 t^2}{2}) \\ &= \exp[t \sum_{i=1}^n \mu_i + \frac{t^2}{2} \sum_{i=1}^n \sigma_i^2] \quad \text{(by the mgf of an independent sum)} \end{aligned}$$

This is the mgf of $N(\sum\mu_i, \sum\sigma_i^2)$, hence the claim (by uniqueness of the mgf). ■

Corollary 3.3.1 — distribution of the sample mean

For a random sample of size n from $N(\mu, \sigma^2)$:

$$\overline{X}_n \sim N(\mu,\ \frac{\sigma^2}{n}) \quad \text{(by Theorem 3.3.1 and the linear transformation property of the normal)}$$

Example 3.3.1 (Ex 3.5) — production time

$X \sim N(6, 2^2)$ independent, $n = 10$; then $\sum X_i \sim N(60{,}40)$ (by Theorem 3.3.1) and

$$P(\sum_{i=1}^{10} X_i \geq 70) = P\left[\frac{\sum X_i - 60}{\sqrt{40}} \geq \frac{70 - 60}{\sqrt{40}}\right] = 1 - \Phi(1.58) = 0.057$$

Theorem 3.3.2 (Thm 3.9) — independence of $\overline{X}_n$ and S_n^2, and the distribution of S_n^2

For a random sample of size n from $N(\mu, \sigma^2)$:

1. $\overline{X}_n$ and S_n^2 are independent
2. $\frac{(n-1)S_n^2}{\sigma^2} \sim \chi^2(n-1)$

Note that the deviations $X_i - \overline{X}_n$ are not independent, because of the constraint $\sum_i (X_i - \overline{X}_n) = 0$; Theorem 3.2.4 therefore cannot be applied to S_n^2 directly.

Proof of (1), by an orthogonal transformation. An orthogonal matrix satisfies $Q^\top Q = QQ^\top = I$ and preserves squared length:

$$\|\|^2 = x^\top Q^\top Q x = x^\top x = \|x\|^2$$

For an iid sample the covariance matrix is $\mathrm{Cov}(X) = \sigma^2 I$, so for $Y = QX$

$$\mathrm{Cov}(Y) = Q\mathrm{Cov}(X)Q^\top = Q(\sigma^2 I)Q^\top = \sigma^2 QQ^\top = \sigma^2 I$$

Y is multivariate normal (a linear transformation of a multivariate normal), and with $\Sigma = \sigma^2 I$ the quadratic form in the exponent separates componentwise, so the joint density factors into its marginals: $Y_1, \ldots, Y_n$ are mutually independent, and functions of disjoint index sets are independent.

Take Q whose first row is $\left(1/\sqrt{n}, \dots, 1/\sqrt{n}\right)$ (a Helmert matrix). Then

$$Y_1 = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i = \sqrt{n}\overline{X}_n$$

$$\sum_{i=2}^{n} Y_i^2 = \sum_{i=1}^{n} X_i^2 - Y_1^2 = \sum_{i=1}^{n} X_i^2 - n\overline{X}_n^2 = \sum_{i=1}^{n} (X_i - \overline{X}_n)^2$$
$$= (n-1)S_n^2 \quad \text{(by length preservation)}$$

Thus $\overline{X}_n = Y_1/\sqrt{n}$ is a function of Y_1 alone and $S_n^2 = \frac{1}{n-1}\sum_{i=2}^{n} Y_i^2$ is a function of $Y_2, \dots, Y_n$ alone; the index sets $\{1\}$ and $\{2, \dots, n\}$ are disjoint, so the two are independent. ■

Proof of (2), by mgf. From the identity $\sum(X_i - \mu)^2 = \sum(X_i - \overline{X}_n)^2 + n(\overline{X}_n - \mu)^2$, dividing by σ^2,

$$V_1 := \sum_{i=1}^{n} \frac{(X_i - \mu)^2}{\sigma^2} = \frac{(n-1)S_n^2}{\sigma^2} + \frac{n(\overline{X}_n - \mu)^2}{\sigma^2}$$

Write $V_2 = \frac{(n-1)S_n^2}{\sigma^2}$ and $V_3 = \frac{n(\overline{X}_n - \mu)^2}{\sigma^2}$, so that $V_1 = V_2 + V_3$.

- $V_1 \sim \chi^2(n)$ (by Theorem 3.2.4)
- $V_3 = ((\overline{X}_n - \mu)/(\sigma/\sqrt{n}))^2 \sim \chi^2(1)$ (by Corollary 3.3.1 and Theorem 3.2.3)
- V_2 is a function of S_n^2 and V_3 of $\overline{X}_n$, so $V_2 \perp V_3$ (by part (1))

$$M_{V_1}(t) = M_{V_2}(t) M_{V_3}(t) \quad \text{(by the mgf of an independent sum)}$$

$$M_{V_2}(t) = \frac{M_{V_1}(t)}{M_{V_3}(t)} = \frac{(1-2t)^{-n/2}}{(1-2t)^{-1/2}} = (1-2t)^{-(n-1)/2}, \quad t < \frac{1}{2} \quad \text{(by Theorem 3.2.1)}$$

This is the mgf of $\chi^2(n-1)$, hence $V_2 \sim \chi^2(n-1)$ (by uniqueness of the mgf). ■

Only the normal distribution keeps its components independent under rotation, so the independence of $\overline{X}_n$ and S_n^2 is a characterization of the normal distribution.

Example 3.3.2 (Ex 3.6) — probability for the sample variance

Continuing Example 3.3.1 ($n = 10$, $\sigma^2 = 4$), $9S_{10}^2/4 \sim \chi^2(9)$ (by Theorem 3.3.2(2)) and

$$P(S_{10}^2 > 5) = P\left[\frac{9S_{10}^2}{4} > \frac{45}{4}\right] = P[\chi^2(9) > 11.25] \approx 0.259$$

Theorem 3.3.3 (Thm 3.10) — ratio of two sample variances

Independent random samples of size n from $N(\mu_X, \sigma_X^2)$ and of size m from $N(\mu_Y, \sigma_Y^2)$:

$$F = \frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F(n-1, m-1)$$

Proof. $(n-1)S_X^2/\sigma_X^2 \sim \chi^2(n-1)$ and $(m-1)S_Y^2/\sigma_Y^2 \sim \chi^2(m-1)$ (by Theorem 3.3.2(2)), and they are independent because the two samples are. Dividing each by its degrees of freedom,

$$F = \frac{[(n-1)S_X^2/\sigma_X^2]/(n-1)}{[(m-1)S_Y^2/\sigma_Y^2]/(m-1)} = \frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \quad \text{(by Definition 3.2.2)} \qquad \blacksquare$$

In particular, if $\sigma_X^2 = \sigma_Y^2$ then $S_X^2/S_Y^2 \sim F(n-1, m-1)$, which is the basis of the test for equality of two variances.

Theorem 3.3.4 (Thm 3.11) — studentized t statistic

For a random sample from $N(\mu, \sigma^2)$, with $S_n = \sqrt{S_n^2}$:

$$T = \frac{\sqrt{n}(\overline{X}_n - \mu)}{\sqrt{\sum_{i=1}^{n}(X_i - \overline{X}_n)^2/(n-1)}} = \frac{\overline{X}_n - \mu}{S_n/\sqrt{n}} \sim t(n-1)$$

Proof.

$$Z = \frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} \sim N(0,1), \qquad U = \frac{(n-1)S_n^2}{\sigma^2} \sim \chi^2(n-1)$$

and Z, U are independent (by Theorem 3.3.2), so

$$\begin{aligned} T = \frac{Z}{\sqrt{U/(n-1)}} &= \frac{\sqrt{n}(\overline{X}_n - \mu)/\sigma}{\sqrt{[(n-1)S_n^2/\sigma^2]/(n-1)}} \\ &= \frac{\sqrt{n}(\overline{X}_n - \mu)}{S_n} \sim t(n-1) \quad \text{(by Definition 3.2.3)} \qquad \blacksquare \end{aligned}$$

Studentization replaces the nuisance parameter σ^2 by S_n^2, thereby eliminating it. As $n \to \infty$ the denominator converges to σ, so $t(n-1) \to N(0,1)$ (by Slutsky's theorem; see Example 3.4.5).

3.4 Law of Large Numbers and Central Limit Theorem

3.4.1 Law of large numbers

Definition 3.4.1 (Def 3.5) — convergence in probability

$$\lim_{n\to\infty} P(|X_n - X| \geq \epsilon) = 0 \ \text{ for every } \epsilon > 0 \qquad \Leftrightarrow \qquad X_n \xrightarrow{p} X$$

equivalently $\lim_{n\to\infty} P(|X_n - X| < \epsilon) = 1$.

Theorem 3.4.1 (Thm 3.12) — law of large numbers

For a random sample from a population with mean $\mu < \infty$:

$$\overline{X}_n \xrightarrow{p} \mu$$

Proof. Assume $\sigma^2 < \infty$. For any $\epsilon > 0$,

$$P[|\overline{X}_n - \mu| < \epsilon] \geq 1 - \frac{E(\overline{X}_n - \mu)^2}{\epsilon^2} = 1 - \frac{\sigma^2/n}{\epsilon^2} \ \longrightarrow \ 1 \quad (n \to \infty)$$

(by Chebyshev's inequality and $E(\overline{X}_n - \mu)^2 = \mathrm{Var}(\overline{X}_n) = \sigma^2/n$, Theorem 3.1.1). ■

Theorem 3.4.2 — convergence in probability of the sample variance

If $E(X_1^4) < \infty$, then

$$S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2 \ \xrightarrow{p} \ \sigma^2$$

Proof. From Theorem 3.1.1, $E(S_n^2) = \sigma^2$ and $\mathrm{Var}(S_n^2) = \frac{1}{n}(\mu_4 - \frac{n-3}{n-1}\sigma^4)$, and $E(X_1^4) < \infty$ means $\mu_4 < \infty$. For any $\epsilon > 0$,

$$P[|S_n^2 - \sigma^2| \geq \epsilon] \leq \frac{\mathrm{Var}(S_n^2)}{\epsilon^2} = \frac{1}{n\epsilon^2}(\mu_4 - \frac{n-3}{n-1}\sigma^4) \ \longrightarrow \ 0 \quad \text{(by Chebyshev's inequality)}$$

since the bracket converges to $\mu_4 - \sigma^4$ while the factor $1/n$ vanishes. ■

Example 3.4.1 (Ex 3.7) — convergence of the sample proportion

For independent Bernoulli(p) variables, $E(X_1) = p$, so

$$\hat{p}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{p} p \quad \text{(by Theorem 3.4.1)}$$

3.4.2 Central limit theorem

Definition 3.4.2 (Def 3.6) — convergence in distribution

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x) \text{ at every continuity point of } F_X \qquad \Leftrightarrow \qquad X_n \xrightarrow{d} X$$

What converges is the cdf, not the random variable, and only at continuity points of F_X.

Theorem 3.4.3 (Thm 3.13) — continuity theorem

If $\lim_{n\to\infty} M_n(t) = M(t)$ on an open interval $-h < t < h$ and $M(t)$ is the mgf of a distribution with cdf F, then $\lim_{n\to\infty} F_n(x) = F(x)$ at every continuity point of F.

For discrete variables the pgf may be used instead: if $\lim_n G_n(s) = G(s)$ for $0 \le s < 1$, then $\lim_n P(X_n = k) = a_k$ where $G(s) = \sum_k a_k\, s^k$.

Theorem 3.4.4 (Thm 3.14) — central limit theorem

For a random sample from a population with mean μ and variance $\sigma^2 < \infty$:

$$Z_n = \frac{\sum_{i=1}^{n} X_i - E(\sum_{i=1}^{n} X_i)}{\sqrt{\mathrm{Var}(\sum_{i=1}^{n} X_i)}} = \frac{\sum_{i=1}^{n}(X_i - \mu)}{\sqrt{n}\sigma} \xrightarrow{d} N(0,1)$$

Lemma 3.4.1 — Lagrange form of the first-order remainder

Let f be twice differentiable on an open interval containing a, and define the remainder as everything the first-order approximation misses:

$$R(x) := f(x) - f(a) - f'(a)(x-a) \qquad \Leftrightarrow \qquad f(x) = f(a) + f'(a)(x-a) + R(x)$$

Then for some ξ between a and x,

$$R(x) = \frac{f''(\xi)}{2}(x-a)^2$$

Proof. Fix $x \neq a$ and define M by $R(x) = \frac{(x-a)^2}{2}M$. Put

$$g(u) = f(u) - f(a) - f'(a)(u-a) - \frac{(u-a)^2}{2}M$$

so that $g(a) = 0$, $g'(a) = 0$, and $g(x) = 0$ (the last by the definition of M).

- $g(a) = g(x) = 0 \Rightarrow g'(\xi_1) = 0$ for some $\xi_1 \in (a, x)$ (by Rolle's theorem)
- $g'(a) = g'(\xi_1) = 0 \Rightarrow g''(\xi_2) = 0$ for some $\xi_2 \in (a, \xi_1)$ (by Rolle's theorem)

Differentiating twice kills the first-order part and leaves M:

$$g''(u) = f''(u) - M, \qquad g''(\xi_2) = f''(\xi_2) - M = 0 \iff M = f''(\xi_2)$$

Writing $\xi = \xi_2$ gives the claim. ■

The point used below is that ξ is trapped between a and x, so $x \to a$ forces $\xi \to a$.

Proof of Theorem 3.4.4 (assuming the mgf exists). (1) The mgf of the centered variable is common to all i:

$$m(t) = E[e^{t(X_i-\mu)}], \qquad m(0) = 1, \quad m'(0) = E(X_i - \mu) = 0, \quad m''(0) = E(X_i-\mu)^2 = \sigma^2$$

(2) Expanding about 0 and splitting off the leading term,

$$m(t) = m(0) + m'(0)t + \frac{m''(\xi)t^2}{2} = 1 + \frac{\sigma^2 t^2}{2} + \frac{[m''(\xi) - \sigma^2]t^2}{2}, \qquad 0 < \xi < t$$

(by Taylor series with the Lagrange remainder, Lemma 3.4.1)

(3) Substituting $t/(\sqrt{n}\sigma)$,

$$M_{Z_n}(t) = \prod_{i=1}^{n} M_{X_i-\mu}\left(\frac{t}{\sqrt{n}\sigma}\right) = \left[m\left(\frac{t}{\sqrt{n}\sigma}\right)\right]^n = \left[1 + \frac{t^2}{2n} + \frac{[m''(\xi) - \sigma^2]t^2}{2n\sigma^2}\right]^n, \quad 0 < \xi < \frac{t}{\sqrt{n}\sigma}$$

(by the mgf of an independent sum)

(4) For fixed t, $t/(\sqrt{n}\sigma) \to 0$ and ξ is squeezed to 0; m'' is continuous at 0, so $m''(\xi) - \sigma^2 \to m''(0) - \sigma^2 = 0$. The bracket is $1 + (t^2/2 + o(1))/n$, hence

$$\lim_{n\to\infty} M_{Z_n}(t) = \exp(t^2/2) \quad \text{(by the exponential limit } (1 + c/n)^n \to e^c\text{)}$$

(5) This is the mgf of $N(0,1)$, so $Z_n \xrightarrow{d} N(0,1)$ (by Theorem 3.4.3 and uniqueness of the mgf). ■

The CLT requires only a finite mean and variance, whatever the shape of the population; it fails for e.g. the Cauchy distribution, which has no mean. Equivalently

$$Z_n = \frac{\overline{X}_n - E(\overline{X}_n)}{\sqrt{\mathrm{Var}(\overline{X}_n)}} = \frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \quad (3.8)$$

so the CLT approximates the distribution of $\overline{X}_n$ or of $\sum X_i$ by a normal distribution.

Example 3.4.2 (Ex 3.8) — normal approximation, continuous case

For a random sample from $U(0,1)$, $E(X_i) = \frac{1}{2}$ and $\mathrm{Var}(X_i) = \frac{1}{12}$, so $\sum X_i \approx N(n/2,\ n/12)$ and

$$P[a \le \sum_{i=1}^{n} X_i \le b] \fallingdotseq \Phi\left[\frac{b - n/2}{\sqrt{n/12}}\right] - \Phi\left[\frac{a - n/2}{\sqrt{n/12}}\right] \quad \text{(by Theorem 3.4.4)}$$

Example 3.4.3 (Ex 3.9) — normal approximation, binomial case

$X \sim B(n,p)$ has the same distribution as $\sum_{i=1}^{n} X_i$ for independent Bernoulli(p) variables, with $E(X_i) = p$ and $\mathrm{Var}(X_i) = pq$, $q = 1 - p$. Hence $(\sum X_i - np)/\sqrt{npq}$ is approximately $N(0,1)$ and

$$P[a \le X \le b] \fallingdotseq \Phi\left[\frac{b - np}{\sqrt{npq}}\right] - \Phi\left[\frac{a - np}{\sqrt{npq}}\right] \quad \text{(by Theorem 3.4.4)}$$

Definition 3.4.3 — continuity correction

$P(X = k)$ is the area of a bar of width 1 centered at k, whereas the normal approximation integrates a curve; taking the integration limits at the integers a, b (the bar centers) cuts off the outer halves of the two end bars. Extending each limit by $\frac{1}{2}$ covers the whole bars:

$$P(a - \frac{1}{2} \ \le \ X \ \le \ b + \frac{1}{2})$$

$$P[a \le X \le b] \doteqdot \Phi\left[\frac{b + 1/2 - np}{\sqrt{npq}}\right] - \Phi\left[\frac{a - 1/2 - np}{\sqrt{npq}}\right]$$

Example 3.4.4 — effect of the continuity correction

$n = 50$, $p = 0.75$, $a = 36$, $b = 37$, so $np = 37.5$ and $npq = 9.375$.

- without correction: $\Phi\left[\frac{37-37.5}{\sqrt{9.375}}\right] - \Phi\left[\frac{36-37.5}{\sqrt{9.375}}\right] = 0.1230$
- with correction: $\Phi\left[\frac{37.5-37.5}{\sqrt{9.375}}\right] - \Phi\left[\frac{35.5-37.5}{\sqrt{9.375}}\right] = 0.2432$
- exact binomial: $P(36 \le X \le 37) = 0.2371$

3.4.3 Slutsky's theorem and the delta method

Theorem 3.4.5 (Thm 3.15) — Slutsky's theorem

If $X_n \overset{p}{\to} c$ for a constant c and $Y_n \overset{d}{\to} Z$, then

1. $Y_n + X_n \overset{d}{\to} Z + c$
2. $X_n Y_n \overset{d}{\to} cZ$

Proof of (1) (outline). For any $\epsilon > 0$, splitting on the event $|X_n - c| < \epsilon$,

$$F_{Y_n+X_n}(z) = P(Y_n + X_n \le z,\ |X_n - c| < \epsilon) + P(Y_n + X_n \le z,\ |X_n - c| \ge \epsilon)$$

$$F_{Y_n+X_n}(z) \le P(Y_n \le z - c + \epsilon) + P(|X_n - c| \ge \epsilon)$$

and the same splitting applied to $P(Y_n + c + \epsilon \le z)$ gives the lower bound, so

$$P(Y_n \le z - c - \epsilon) - P(|X_n - c| \ge \epsilon) \ \le \ F_{Y_n+X_n}(z) \ \le \ P(Y_n \le z - c + \epsilon) + P(|X_n - c| \ge \epsilon)$$

This holds for every $\epsilon > 0$, and both bounds converge to $P(Z \le z - c) = P(Z + c \le z)$ wherever Z is continuous at $z - c$ (by $Y_n \overset{d}{\to} Z$ and $X_n \overset{p}{\to} c$). ■

Example 3.4.5 (Ex 3.10) — limiting distribution of the studentized statistic

For a random sample from a population with mean μ and variance $\sigma^2 < \infty$ (no normality assumed): $S_n^2 \xrightarrow{p} \sigma^2$ (by Theorem 3.4.2), hence $S_n \xrightarrow{p} \sigma$ (since $\sqrt{\cdot}$ is continuous), and $\left(\overline{X}_n - \mu)/(\sigma/\sqrt{n}) \xrightarrow{d} N(0,1)\right)$ (by Theorem 3.4.4). Therefore

$$\frac{\overline{X}_n - \mu}{S_n/\sqrt{n}} = \frac{\sigma}{S_n} \cdot \frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1) \quad \text{(by Theorem 3.4.5(2))}$$

Combined with Theorem 3.3.4 this confirms that $t(n-1) \to N(0,1)$ as the degrees of freedom grow.

Theorem 3.4.6 (Thm 3.16) — delta method

If $\sqrt{n}(X_n - \theta) \xrightarrow{d} N(0, \sigma^2)$ and g' is continuous and nonzero at θ, then

$$\sqrt{n}(g(X_n) - g(\theta)) \xrightarrow{d} N(0,\ \sigma^2[g'(\theta)]^2)$$

Proof. For some $\tilde{\theta}$ between X_n and θ,

$$g'(\tilde{\theta}) = \frac{g(X_n) - g(\theta)}{X_n - \theta} \qquad \Leftrightarrow \qquad g(X_n)$$
$$= g'(\tilde{\theta})(X_n - \theta) + g(\theta) \quad \text{(by the mean value theorem)}$$

$$X_n - \theta = \frac{1}{\sqrt{n}} \cdot \sqrt{n}(X_n - \theta) \xrightarrow{d} 0 \cdot N(0, \sigma^2) = 0 \quad \text{(by Theorem 3.4.5(2))}$$

Convergence in distribution to a constant is convergence in probability, so $X_n \xrightarrow{p} \theta$; $\tilde{\theta}$ is trapped between X_n and θ, so $\tilde{\theta} \xrightarrow{p} \theta$ (by the squeeze theorem), and $g'(\tilde{\theta}) \xrightarrow{p} g'(\theta)$ since g' is continuous. Hence

$$\sqrt{n}(g(X_n) - g(\theta)) = g'(\tilde{\theta}) \cdot \sqrt{n}(X_n - \theta) \xrightarrow{d} g'(\theta)N(0, \sigma^2) = N(0, \sigma^2[g'(\theta)]^2)$$

(by Theorem 3.4.5(2)) ■

Example 3.4.6 (Ex 3.11) — variance stabilizing transformation

For a random sample from Poisson(λ), $E(X_i) = \mathrm{Var}(X_i) = \lambda$, so

$$Z_n = \sqrt{n}(\overline{X}_n - \lambda) \xrightarrow{d} N(0, \lambda) \quad \text{(by Theorem 3.4.4)}$$

With $g(x) = \sqrt{x}$, $g'(\lambda) = \frac{1}{2\sqrt{\lambda}}$ and $\left[g'(\lambda)\right]^2 = \frac{1}{4\lambda}$, so

$$\sqrt{n}(\sqrt{\overline{X}_n} - \sqrt{\lambda}) \xrightarrow{d} N(0, \frac{1}{4}) \quad \text{(by Theorem 3.4.6)}$$

The limiting variance no longer depends on λ: this is a variance stabilizing transformation.

Theorem 3.4.7 — multivariate delta method: approximate mean and variance

For a random vector $X = (X_1, \dots, X_k)$ with $E(X) = \theta = (\theta_1, \dots, \theta_k)$,

$$g(x) \approx g(\theta) + \sum_{i=1}^{k} g_i'(\theta)(x_i - \theta_i), \qquad g_i'(\theta) = \frac{\partial}{\partial x_i} g(x)|_{x=\theta} \quad \text{(by Taylor series)}$$

$$Eg(X) \approx g(\theta) + \sum_{i=1}^{k} g_i'(\theta) E(X_i - \theta_i) = g(\theta) \quad (\text{by } E(X_i - \theta_i) = 0) \quad (3.9)$$

$$\mathrm{Var} g(X) \approx \sum_{i=1}^{k} [g_i'(\theta)]^2 \mathrm{Var}(X_i) + 2 \sum_{i>j} g_i'(\theta) g_j'(\theta) \mathrm{Cov}(X_i, X_j) \quad (3.10)$$

(3.10) follows by expanding $\mathrm{Var} g(X) \approx E[(\sum_i g_i'(\theta)(X_i - \theta_i))^2]$.

Example 3.4.7 (Ex 3.12) — ratio-type estimator

Estimating $g(\mu_X, \mu_Y) = \frac{\mu_X}{\mu_Y}$ by $\frac{\overline{X}_n}{\overline{Y}_m}$, the partial derivatives are

$$\frac{\partial}{\partial \mu_X} g = \frac{1}{\mu_Y}, \qquad \frac{\partial}{\partial \mu_Y} g = -\frac{\mu_X}{\mu_Y^2}$$

$$E\left(\frac{\overline{X}_n}{\overline{Y}_m}\right) \approx \frac{\mu_X}{\mu_Y} \quad \text{(by (3.9))}$$

$$\mathrm{Var}\left(\frac{\overline{X}_n}{\overline{Y}_m}\right) \approx \frac{1}{\mu_Y^2} \mathrm{Var}(\overline{X}_n) + \frac{\mu_X^2}{\mu_Y^4} \mathrm{Var}(\overline{Y}_m) - 2\frac{\mu_X}{\mu_Y^3} \mathrm{Cov}(\overline{X}_n, \overline{Y}_m) \quad \text{(by (3.10))}$$

$$= \left(\frac{\mu_X}{\mu_Y}\right)^2 \left(\frac{\mathrm{Var}(\overline{X}_n)}{\mu_X^2} + \frac{\mathrm{Var}(\overline{Y}_m)}{\mu_Y^2} - 2\frac{\mathrm{Cov}(\overline{X}_n, \overline{Y}_m)}{\mu_X \mu_Y}\right)$$

3.5 Order Statistics

Definition 3.5.1 — order statistics

The random sample $X_1, \ldots, X_n$ arranged in increasing order:

$$X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)} \quad (3.11)$$

Theorem 3.5.1 — joint density of the order statistics

For continuous f,

$$g(x_{(1)}, \ldots, x_{(n)}) = \begin{cases} n!\, f(x_{(1)}) f(x_{(2)}) \cdots f(x_{(n)}), & x_{(1)} < x_{(2)} < \cdots < x_{(n)} \\ \text{[4pt]}0, & \text{otherwise} \end{cases} \quad (3.12)$$

Proof. Sorting is not one-to-one: $n!$ different samples map to the same ordered vector, e.g. all $3! = 6$ permutations of $(5, 1, 3)$ map to $(1, 3, 5)$, so the change-of-variable formula $f_Y(y) = f_X(g^{-1}(y))|J|$ cannot be applied directly.

Partition the sample space into the $n!$ disjoint regions on which the ordering is fixed, e.g. $A_1 = \{x_1 < x_2 < \cdots < x_n\}$; on each region sorting is one-to-one and merely permutes coordinates, so $|J_i| = 1$ and the density there is $f(x_{(1)}) \cdots f(x_{(n)})$.

For an iid sample the density is invariant under permutation of its arguments, so all $n!$ regions contribute equally; summing them multiplies one region's density by $n!$, which gives (3.12). The boundaries $x_i = x_j$ have probability 0 in the continuous case, so strict and weak inequalities give the same density. ■

Example 3.5.1 (Ex 3.13) — joint density for $f(x) = 3x^2$

X_1, X_2, X_3 a random sample from $f(x) = 3x^2$, $0 < x < 1$; for $0 < x_{(1)} < x_{(2)} < x_{(3)} < 1$,

$$g(x_{(1)}, x_{(2)}, x_{(3)}) = 3!\,(3x_{(1)}^2)(3x_{(2)}^2)(3x_{(3)}^2) = 162 x_{(1)}^2 x_{(2)}^2 x_{(3)}^2$$

and 0 otherwise.

Theorem 3.5.2 (Thm 3.17) — density of the k-th order statistic

For a population with density f and cdf F, with $f(x) > 0$ on $a < x < b$:

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!\,(n-k)!}[F(x)]^{k-1}[1-F(x)]^{n-k}f(x), \qquad a < x < b$$

and 0 otherwise.

Proof. $X_{(k)} = x$ requires one observation at x (density $f(x)$), $k-1$ observations below x (each with probability $P[X \le x] = F(x)$), and $n-k$ observations above x (each with probability $P[X > x] = 1 - F(x)$). The number of such arrangements is $\frac{n!}{(k-1)!(n-k)!}$, since the order within the low group and within the high group is irrelevant. Multiplying gives the density. ■

Corollary 3.5.1 — joint density of $X_{(i)}$ and $X_{(j)}$, $i < j$

By the same argument, for $x_{(i)} < x_{(j)}$,

$$\begin{aligned} f_{X_{(i)},X_{(j)}}(x_{(i)}, x_{(j)}) \\ &= \frac{n!}{(i-1)!\,(j-i-1)!\,(n-j)!}[F(x_{(i)})]^{i-1}f(x_{(i)})[F(x_{(j)}) \\ &- F(x_{(i)})]^{j-i-1}f(x_{(j)})[1-F(x_{(j)})]^{n-j} \end{aligned}$$

Theorem 3.5.3 — sample minimum and sample maximum

The cdfs follow directly from the definition:

$$G_1(x) = P[X_{(1)} \le x] = 1 - P[\text{all } X_i > x] = 1 - [1-F(x)]^n$$

$$G_n(x) = P[X_{(n)} \le x] = P[\text{all } X_i \le x] = [F(x)]^n$$

Differentiating in the continuous case,

$$f_{X_{(1)}}(x) = \frac{d}{dx}G_1(x) = n[1-F(x)]^{n-1}f(x) \quad (3.13)$$

$$f_{X_{(n)}}(x) = \frac{d}{dx}G_n(x) = n[F(x)]^{n-1}f(x) \quad (3.14)$$

Example 3.5.2 (Ex 3.14) — minimum of an exponential sample

For a random sample from $f(x) = \frac{1}{\lambda}e^{-x/\lambda}$, $x > 0$, we have $F(x) = 1 - e^{-x/\lambda}$, so

$$f_{X_{(1)}}(x) = n[1-(1-e^{-x/\lambda})]^{n-1}\frac{1}{\lambda}e^{-x/\lambda} = \frac{n}{\lambda}\exp(-\frac{nx}{\lambda}), \qquad x > 0 \quad \text{(by (3.13))}$$

i.e. $X_{(1)}$ is exponential with mean λ/n.

Definition 3.5.2 — sample median, sample range

The sample median is the middle order statistic, and measures location:

- n odd: $\tilde{X} = X_{(k)}, \quad k = \frac{n+1}{2}$
- n even: $\tilde{X} = \frac{X_{(k)}+X_{(k+1)}}{2}, \quad k = \frac{n}{2}$

The sample range measures dispersion:

$$R = X_{(n)} - X_{(1)}$$